CALCULUS

Chapter 14

Limits & Derivatives

14

SYLLABUS 5.1–5.4 5.6 5.7 5.12 5.13 5.15

A car's speedometer reads 80 km/h. Ask what that number means, and you meet a problem that mathematicians could not solve for two thousand years.

Speed is distance divided by time. But the speedometer shows your speed *right now*, at a single instant. In one instant no time passes. No distance is covered either. So speed at an instant appears to be $0 \div 0$, which has no value. How can the needle point to 80?

The way out is to stop asking about the instant. Ask about short intervals around it instead.

Measure your average speed over the next hour. Then over the next minute. Then the next second, then the next hundredth of a second. Each interval is shorter than the one before, and each average is closer to one particular number. That number is the speed at the instant. The averages reach it, even though the instant alone cannot give it.

This idea is called the **limit**. You cannot reach a value directly, so you find it by seeing what nearby values approach. All of calculus is built on it.

From the limit comes the **derivative**. It measures the exact rate at which something changes: how steep a curve is at one point, or how fast an apple is falling at the moment it lands. This chapter builds the derivative from the limit, and then develops methods for calculating it quickly.

By the end of this chapter you will be able to evaluate limits, say when a function is continuous and differentiable, and differentiate from first principles. You will apply the product, quotient and chain rules, differentiate the standard functions of the course, and find the equation of a tangent and a normal. Calculus begins here because a derivative answers two questions at once. Geometrically it is the gradient of a curve at a point; physically it is a rate of change. Every later application is one of those two readings.

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14.1 Limits

A **limit** is the value a function approaches as its input approaches some point. It does not matter what the function does at that point itself. We write

$$\lim_{x \to a} f(x) = L$$

to mean: as x gets **arbitrarily close** (as close as you like) to a, $f(x)$ gets arbitrarily close to L. The words "whatever happens at a" matter. A limit can exist at a point where the function itself is undefined.

For most functions at most points, the limit is found by direct substitution. If f is defined at a and its graph passes through that point unbroken, then the value it approaches is simply the value it takes:

$$\lim_{x \to 2} (x^2 + 3x) = 2^2 + 3(2) = 10.$$

That case is straightforward. The useful cases are the ones where substitution fails.

EXAMPLE 14.1
Find $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$.
Substituting $x = 3$ gives $\frac{0}{0}$, which is undefined: the function has a hole at $x = 3$. But for every $x \neq 3$ we can factor and cancel:

$$\frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x-3} = x + 3.$$

Away from the hole the function is simply $x + 3$, so as $x \to 3$ it approaches 6:

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = 6.$$

At $x = 3$ the function has no value at all, yet 6 is the value it approaches. The difference between "approaches" and "equals" is what a limit measures.

The form $\frac{0}{0}$ is called **indeterminate**. It does not mean the limit fails to exist. It means more work is needed.

One limit of this kind is worth memorising, because the derivatives of the trigonometric functions rest on it:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

When there is no algebraic method, you can *estimate* a limit. Calculate the function at values closer and closer to the target, from both sides. Here is the result for this one, with x in radians:

x	-0.1	-0.01	0.01	0.1
$\dfrac{\sin x}{x}$	0.99833	0.99998	0.99998	0.99833

Both sides approach 1. A table or a zoomed-in graph is a fair way to **conjecture** (guess, before proving) the value of a limit, and on a calculator paper it is often the fastest way to check an algebraic answer.

Reading a limit off the graph

A graph often shows why a limit takes the value it does, and the two fractions $\dfrac{\sin x}{x}$ and $\dfrac{\cos x}{x}$ make the contrast clearly. Both have denominator $x \to 0$, yet they behave completely differently, and the reason is what the *numerator* does at the same moment.

Sketch $y = \sin x$ and $y = x$ on the same axes. Near the origin the two graphs are almost indistinguishable: at $x = 0.1$, $\sin x = 0.0998$. Numerator and denominator shrink to zero together, at the same rate, so their ratio approaches 1.

Now sketch $y = \cos x$ against $y = x$. As $x \to 0$ the numerator approaches 1 while the denominator approaches 0. A fixed quantity divided by something ever smaller grows without bound, so the fraction does not approach any finite value.

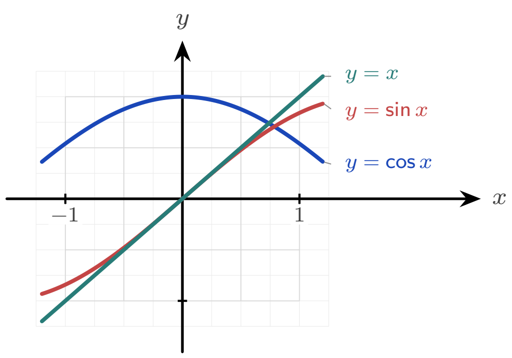

Figure 14.1 Near the origin the graphs of $\sin x$ and x are almost the same, so their ratio tends to 1. But $\cos x$ tends to 1 while x tends to 0, so $\frac{\cos x}{x}$ has no finite limit.

CAUTION Be precise about the second case. As $x \to 0^+$ the fraction $\frac{\cos x}{x}$ increases without bound, but as $x \to 0^-$ it decreases without bound, because x is negative there. The two sides disagree, so $\lim_{x\to 0} \frac{\cos x}{x}$ does not exist. Writing that it ``equals infinity'' conceals that disagreement, so it is not an acceptable answer.

Not every limit exists. A limit **converges** when the values approach one finite number, and **diverges** when they do not: $\frac{1}{x^2}$ grows without bound as $x \to 0$, and $\sin x$ oscillates between -1 and 1 as $x \to \infty$ without approaching any value. You have met this distinction before: the infinite geometric series of Ch. 2 converges exactly when $|r| < 1$, because its partial sums approach a single value, and diverges otherwise.

CAUTION A limit can fail to exist. If $f(x)$ approaches different values from the left and the right of a, as at a jump, there is no single limit. ``Approaches $\frac{0}{0}$'' is fixable; ``approaches two different things'' is not.

YOUR TURN 14.1 Limits

1. Find each limit by direct substitution.

(a) $\lim_{x\to 2}(3x-1)$ [1]

(b) $\lim_{x\to 0}(x^2+2x+5)$ [1]

(c) $\lim_{x\to 3}(x^2-9)$ [1]

(d) $\lim_{x\to 1}\dfrac{x+1}{x+3}$ [2]

2. Each of these gives $\frac{0}{0}$. Factorise, cancel, then take the limit.

(a) $\lim_{x\to 4}\dfrac{x^2-16}{x-4}$ [2]

(b) $\lim_{x\to 1}\dfrac{x^2-1}{x-1}$ [2]

(c) $\lim_{x\to 2}\dfrac{x^2-x-2}{x-2}$ [2]

(d) $\lim_{x\to -3}\dfrac{x^2-9}{x+3}$ [2]

3. Write down the value of $\lim_{x\to 0}\dfrac{\sin x}{x}$. [1]

4. Explain why $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist. [2]

5. State whether each of the following converges or diverges as $x \to \infty$, and give the limit where one exists.

(a) $\dfrac{1}{x}$ [1]

(b) $\sin x$ [1]

(c) x^2 [1]

(d) $\dfrac{3x+1}{x}$ [2]

6. GDC Estimate $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$ by evaluating the expression at $x = 0.1$ and $x = 0.01$ radians. State the exact value you think the limit takes. [3]

14.2 The Derivative

Calculus is the mathematics of change. You already know how to measure change: it is the gradient.

Rise over run gives the change in y for each unit of change in x. If the gradient doubles, the quantity changes twice as fast.

There is one problem. A gradient is defined for a *straight line* only. One number describes the whole line, because a line changes at the same rate everywhere.

Very few real quantities behave like that. A cooling cup, a thrown ball, a growing population: in each case the rate of change is itself changing. So the question "what is the gradient of this curve?" has no single answer. The answer is different at every point.

So ask a smaller question. Do not ask for the gradient of the curve. Ask for the gradient of the curve *at one point*. Once you can answer that for one point, you can answer it for every point, and the rate of change becomes a function of its own.

Here is the method. On the graph of $y = f(x)$, choose your point x and a second point $x + h$ close to it. Join the two points with a straight line, called the **secant**, and find its gradient. You can already calculate that:

$$\frac{f(x+h) - f(x)}{h},$$

the rise over the run. This is the average rate of change over the interval of width h.

The secant gradient is still not the answer. It is an average over an interval of width h, and the question was about a single point.

Now make h smaller and smaller. The second point moves towards the first, and the secant rotates towards a limiting position, meeting the curve at one point only. That line is the **tangent**. Its gradient is the rate of change at a single point, and it is called the **derivative**.

That is the complete idea. A gradient is defined only for a line, so we find the one line that matches the curve at that point, and use its gradient instead.

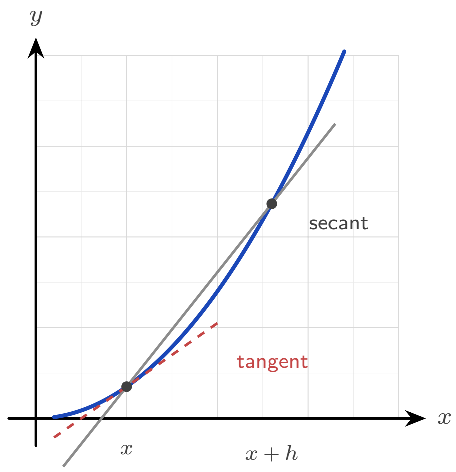

Figure 14.2 The secant joins two points on the curve and measures an average rate of change. As $h \to 0$ the second point moves to the first and the secant approaches the tangent.

DEF · IB The **derivative** of f at x, written $f'(x)$, is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

when this limit exists. Computing a derivative directly from this definition is called differentiation **from first principles**.

The derivative is itself a function. Give it a value of x and it gives back the gradient of the curve at that value. Three notations mean the same thing: $f'(x)$, $\frac{dy}{dx}$ and $\frac{d}{dx}f(x)$.

EXAMPLE 14.2

Differentiate $f(x) = x^2$ from first principles.

Form the difference quotient and expand:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Now let $h \to 0$:

$$f'(x) = \lim_{h \to 0}(2x + h) = 2x.$$

You must cancel the h before taking the limit. While $h \neq 0$ the division is allowed. Only after cancelling can h be set to zero safely.

EXAMPLE 14.3

Differentiate $f(x) = \frac{1}{x}$ from first principles.

$$\frac{f(x+h) - f(x)}{h} = \frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \frac{1}{h} \cdot \frac{x - (x+h)}{x(x+h)} = \frac{-h}{h\,x(x+h)} = \frac{-1}{x(x+h)}.$$

As $h \to 0$:

$$f'(x) = \lim_{h \to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

TIP In examinations, first-principles differentiation is asked only for polynomials. The $\frac{1}{x}$ example above shows how far the definition extends; it is not itself an examination task.

Continuity and differentiability HL

The derivative is a limit, so it can fail to exist. There are two ways this matters.

A function is **continuous** at a point when its limit there exists and equals its value. Informally: the graph passes through the point with no hole and no jump, and you could draw it without lifting the pen.

A function is **differentiable** at a point when the first-principles limit exists. This is stronger.

The graph must be unbroken and also *smooth*, with one clear tangent at that point.

Every differentiable function is continuous, but the reverse is not true. The standard example is the absolute value.

EXAMPLE 14.4

Show that $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there.
Continuity is clear: as $x \to 0$ from either side, $|x| \to 0 = f(0)$. The graph is an unbroken V.
For differentiability, form the difference quotient at 0:

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The quotient approaches 1 from the right and -1 from the left. The two sides disagree, so the limit does not exist and f has no derivative at 0. On the graph, the V has a *corner*, and a corner has no single tangent. A graph can be unbroken and still fail to be smooth.

TIP Examinations will not ask you to *test* a function for continuity or differentiability. What they expect is the idea itself: you may differentiate only where the curve is smooth.

YOUR TURN 14.2 The Derivative

7. Differentiate each from first principles, showing the limit.

(a) $f(x) = x^2$ [3] (b) $f(x) = 3x^2$ [3]
(c) $f(x) = x^2 + x$ [3] (d) $f(x) = x^2 - 3x$ [3]

8. The definition of the derivative can also be used numerically. For $f(x) = \sin x$, evaluate $\dfrac{f(0+h) - f(0)}{h}$ for $h = 0.1$, $h = 0.01$ and $h = 0.001$, working in radians. State the value the quotient is approaching, and name the standard derivative it confirms. [4]

9. GDC Repeat the same calculation for $f(x) = e^x$ at $x = 0$, then at $x = 1$. Explain what the two answers suggest about the derivative of e^x. [4]

10. Write down the three notations used in this book for the derivative of $y = f(x)$. [2]

11. Explain why $f(x) = |x|$ has no derivative at $x = 0$, even though its graph is unbroken there. [3]

14.3 Differentiating Powers

First principles works, but it is slow. The two worked examples above show a pattern: x^2 gave $2x$, and $\frac{1}{x} = x^{-1}$ gave $-x^{-2}$.

In both cases two things happened. The power decreased by one, and the original power became a multiplier in front. This is the **power rule**.

KEY · IB

$$\frac{d}{dx} x^n = nx^{n-1}$$

At Standard Level this is used with integer powers. It holds just as well for every rational power, which is the form needed from §14.5 onwards and the reason the next example can differentiate $\sqrt{x}$.

Two more rules let you differentiate any polynomial. A constant multiplier is unchanged, and a sum is differentiated one term at a time:

$$\frac{d}{dx}(c\,f(x)) = c\,f'(x), \qquad \frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x).$$

A constant on its own has derivative zero, because its graph is a horizontal line with gradient zero.

EXAMPLE 14.5

Differentiate $y = 3x^4 - 2x^2 + 7$.

Differentiate each term with the power rule:

$$\frac{dy}{dx} = 3(4x^3) - 2(2x) + 0 = 12x^3 - 4x.$$

The constant 7 has derivative zero, as every constant does.

EXAMPLE 14.6

Differentiate $y = \sqrt{x} + \dfrac{1}{x^2}$.

Rewrite both terms as powers before differentiating:

$$y = x^{1/2} + x^{-2} \implies \frac{dy}{dx} = \tfrac{1}{2}x^{-1/2} - 2x^{-3} = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}.$$

The power rule requires the function in the form x^n. Convert every root and every reciprocal to a power before differentiating, and the rest of the question becomes routine.

EXAMPLE 14.7

Find the gradient of $y = x^3 - 4x$ at the point where $x = 2$.

Differentiate, then substitute:

$$\frac{dy}{dx} = 3x^2 - 4, \qquad \left.\frac{dy}{dx}\right|_{x=2} = 3(4) - 4 = 8.$$

The derivative is a function. A gradient *at a point* is that function evaluated at the point.

YOUR TURN 14.3 Differentiating Powers

12. Differentiate each of the following.

(a) $y = x^7$ [1]
(b) $y = 5x^3 - 2x$ [2]
(c) $y = x^4 - 3x^2 + 6$ [2]
(d) $y = 4x^5 + x$ [2]
(e) $y = 2x^3 - 7x^2 + x - 9$ [2]
(f) $y = 6$ [1]

13. Rewrite each function as a power of x, then differentiate. In part (f), divide through first.

(a) $y = 2/x$ [2]
(b) $y = \sqrt{x}$ [2]
(c) $y = x^3 + 1/x$ [2]
(d) $y = 1/x^3$ [2]
(e) $y = \sqrt[3]{x}$ [2]
(f) $y = (x^2 + 1)/x$ [3]

14. Find the gradient of each curve at the point given.

(a) $y = x^2 - 5x$ at $x = 3$ [2]
(b) $y = x^3 - 4x$ at $x = -1$ [2]
(c) $y = \sqrt{x}$ at $x = 4$ [2]
(d) $y = 2x^3 + x$ at $x = 1$ [2]

15. Find the value or values of x at which each curve has the stated gradient.

(a) $y = x^2 - 6x$, gradient 0 [2]
(b) $y = x^3$, gradient 12 [3]

16. Explain why $y = \sqrt[4]{x}$ must be rewritten before the power rule can be applied, and state the rewritten form. [2]

14.4 Tangents and Normals

The derivative gives the gradient of the tangent at any point, and a gradient plus a point determines a line. So you can now write the equation of the **tangent** to a curve at a given point. The **normal** is the line perpendicular to the tangent there, with gradient the negative reciprocal.

The tangent and the normal are both straight lines, so both are written with the same equation, the point-gradient form from Ch. 4:

$$y - y_1 = m(x - x_1),$$

where (x_1, y_1) is a known point on the line and m is its gradient. Only two things ever change between the two lines. The point (x_1, y_1) is the same for both, because the tangent and the normal meet the curve at the same place, so all that differs is the gradient m.

> KEY · IB At the point (x_1, y_1) on the curve $y = f(x)$, both lines have the form $y - y_1 = m(x - x_1)$, with
>
> $$m_{\text{tangent}} = f'(x_1), \qquad m_{\text{normal}} = -\frac{1}{f'(x_1)}.$$

So every question of this type follows three steps: find y_1 by substituting x_1 into f, find $f'(x_1)$ for the gradient, then put both into the point-gradient form.

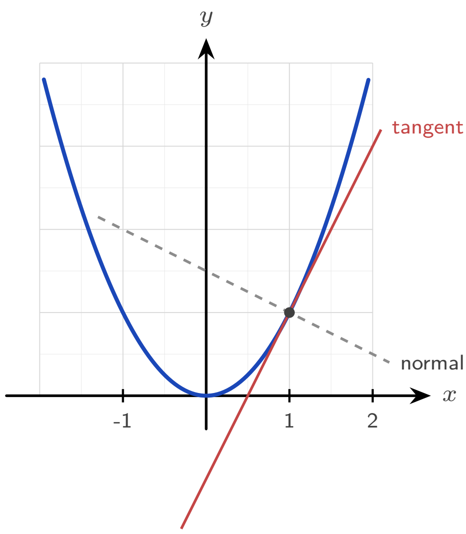

Figure 14.3 At a point on a curve the tangent has gradient $f'(x)$, and the normal is perpendicular to it, with gradient $-1/f'(x)$.

EXAMPLE 14.8

Find the equations of the tangent and normal to $y = x^2 - 3x + 2$ at the point where $x = 2$.

First the point: $y(2) = 4 - 6 + 2 = 0$, so the point is $(2, 0)$.
Then the gradient: $\dfrac{dy}{dx} = 2x - 3$, so at $x = 2$ the tangent gradient is $2(2) - 3 = 1$.
Tangent, through $(2, 0)$ with gradient 1:

$$y - 0 = 1(x - 2) \implies y = x - 2.$$

The normal is perpendicular, so its gradient is -1:

$$y - 0 = -1(x - 2) \implies y = -x + 2.$$

Always find the y-coordinate first. A common error is to write the equation of the line using the x-value alone.

YOUR TURN 14.4 Tangents and Normals

17. Find the equation of the tangent to each curve at the point given.

(a) $y = x^2$ at $(2, 4)$ [2]
(b) $y = x^2 - 4x + 3$ at $x = 1$ [3]
(c) $y = 1/x$ at $x = 1$ [3]
(d) $y = e^x$ at $x = 0$ [3]

18. Find the equation of the normal to each curve at the point given.

(a) $y = x^3$ at $(1, 1)$ [3]
(b) $y = \sqrt{x}$ at $x = 4$ [3]

19. A tangent and a normal are drawn at the same point of a curve. Explain the relationship between their gradients, and state what happens when the tangent is horizontal. [3]

20. The tangent to $y = x^2$ at the point where $x = a$ passes through $(0, -4)$. Find the two possible values of a. [4]

21. Find the coordinates of the point on $y = x^2 - 6x$ at which the tangent is horizontal. [3]

14.5 The Standard Derivatives

The power rule covers algebraic functions. The trigonometric, exponential and logarithmic functions have derivatives of their own. Each can be proved from first principles. You can also find each one by reading gradients from a graph, and it is worth doing that once before you memorise anything.

The method is always the same. Move along the graph of f from left to right. At each point, ask how steep the graph is, and plot that steepness as a height. The curve you plot is f'.

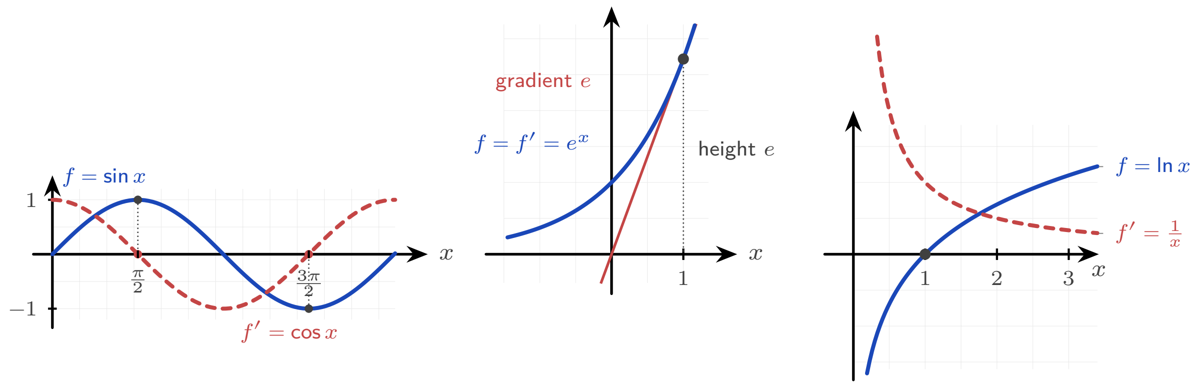

Figure 14.4 Each derivative read from the gradient of its own graph. **Left:** where $\sin x$ is flat, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$, the curve $\cos x$ crosses zero. **Centre:** the tangent to e^x at $x = 1$ has gradient e, which is also the height of the curve there. **Right:** $\ln x$ is steep near 0 and flattens as x grows, and $\frac{1}{x}$ does exactly that.

Read each panel separately.

Sine. At $x = 0$ the sine graph is rising most steeply, so f' starts at its maximum. At the top of the curve, $x = \frac{\pi}{2}$, the graph is flat for an instant, so f' is zero. Past the crest it falls, so f' goes negative. A curve that starts at 1, drops through zero at $\frac{\pi}{2}$ and reaches -1 at π is exactly $\cos x$.

The exponential. Measure the gradient of $y = e^x$ at any point and you get the height of the curve at that point. That is the defining property of e, so here $f' = f$.

The logarithm. Just to the right of 0 the graph of $\ln x$ is almost vertical, so f' is very large. As x increases the curve becomes flatter, so f' decreases towards zero. It stays positive everywhere. That describes $\dfrac{1}{x}$.

Those are the four standard derivatives to learn.

KEY · IB

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}$$

The exponential is the one to look at twice: e^x is its own derivative. More precisely, the functions equal to their own derivative are exactly those of the form $y = Ce^x$, and e^x is the one passing through $(0, 1)$. That property is what makes e important. Note also that the results for sin and cos require x in *radians*.

At Higher Level a wider set of standard derivatives is provided. HL They are listed here for reference and used freely in the rules that follow. The reciprocal and inverse trigonometric functions in the table are defined in Ch. 9; only their derivatives are needed here.

$$\frac{d}{dx}\tan x = \sec^2 x$$
$$\frac{d}{dx}\sec x = \sec x \tan x$$
$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x$$
$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$$

KEY · IB

$$\frac{d}{dx}a^x = a^x \ln a$$
$$\frac{d}{dx}\log_a x = \frac{1}{x \ln a}$$
$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Notice one pair. The derivative of arccos is the negative of the derivative of arcsin, because the two graphs are reflections of each other.

EXAMPLE 14.9

Differentiate $y = 3\sin x + 2e^x - 5\ln x$.

Apply the standard derivatives term by term:

$$\frac{dy}{dx} = 3\cos x + 2e^x - \frac{5}{x}.$$

Constant multipliers are unchanged, exactly as with powers.

YOUR TURN 14.5 The Standard Derivatives

22. Differentiate each of the following.

(a) $y = \sin x + \cos x$ [1] (b) $y = e^x - 3\ln x$ [2]
(c) $y = 4\cos x$ [1] (d) $y = \tan x$ [1]
(e) $y = \arctan x$ [1] (f) $y = 3^x$ [2]

23. Differentiate each of the following. All are in the Higher Level list.

(a) $y = 2\sin x - 5e^x$ [2] (b) $y = \ln x + \tan x$ [2]
(c) $y = \sec x$ [1] (d) $y = \arcsin x$ [1]

24. Find the gradient of $y = \sin x$ at $x = \frac{\pi}{3}$, giving an exact answer. [2]

25. Find the value of x at which the curve $y = e^x$ has gradient 1, and state the significance of that point. [3]

26. The results $\frac{d}{dx}\sin x = \cos x$ and $\frac{d}{dx}\cos x = -\sin x$ hold only when x is measured in radians. Explain why the choice of unit matters. [2]

14.6 The Rules of Differentiation

Most functions are built from simpler ones, by composing, multiplying or dividing them. Three rules cover these three cases. With the standard derivatives, they let you differentiate almost any function in the course.

The chain rule

A **composite** function is a function inside a function, such as $(3x^2 + 1)^5$ or $\sin(2x)$. The **chain rule** differentiates it by working from the outside in: differentiate the outer function, then multiply by the derivative of the inner.

KEY · IB If $y = g(u)$ where $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

The chain rule appears in most differentiation questions once the functions stop being simple powers. Two habits make it reliable:

1. **Identify the inner function first**, before differentiating anything. Ask which part is wrapped inside the other.
2. **Multiply by the derivative of the inner function.** Differentiating the outer function alone and stopping there is a common error, and it is worth checking for at the end of every chain rule question.

EXAMPLE 14.10

Differentiate (a) $y = (3x^2+1)^5$ and (b) $y = e^{x^2}$.

(a) Outer is u^5, inner is $u = 3x^2 + 1$:

$$\frac{dy}{dx} = 5(3x^2+1)^4 \cdot 6x = 30x(3x^2+1)^4.$$

(b) Outer is e^u, inner is $u = x^2$:

$$\frac{dy}{dx} = e^{x^2} \cdot 2x = 2x\, e^{x^2}.$$

In each case the second factor, $6x$ and $2x$, is the derivative of the inner function.

The product rule

To differentiate a product of two functions, you cannot simply multiply their derivatives. It is worth seeing why, because the mistake is a natural one to make.

Take $u = x$ and $v = x$, so that $y = uv = x^2$. The derivative of x^2 is $2x$. But $u' = 1$ and $v' = 1$, and $u'v' = 1$. The two answers do not agree, so multiplying derivatives is simply wrong.

The reason is that a product changes for two separate reasons at the same time.

Suppose both u and v increase. Part of the increase in uv comes from the change in u, multiplied by the whole of v. The rest comes from the change in v, multiplied by the whole of u. The correct rule adds these two parts, which is why each term keeps one factor undifferentiated.

KEY · IB If $y = uv$, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

EXAMPLE 14.11

Differentiate $y = x^2e^x$.

Let $u = x^2$ and $v = e^x$, so $u' = 2x$ and $v' = e^x$:

$$\frac{dy}{dx} = x^2e^x + e^x(2x) = e^x(x^2 + 2x).$$

Factorise the answer where you can. A factorised form makes later work much easier, for example finding where $\frac{dy}{dx} = 0$.

The quotient rule

For a quotient of two functions, the rule is similar but the order of terms matters, since subtraction is involved.

KEY · IB If $y = \dfrac{u}{v}$, then

$$\frac{dy}{dx} = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$$

EXAMPLE 14.12

Differentiate $y = \dfrac{x}{x+1}$.

Let $u = x$ and $v = x + 1$, so $u' = 1$ and $v' = 1$:

$$\frac{dy}{dx} = \frac{(x+1)(1) - x(1)}{(x+1)^2} = \frac{1}{(x+1)^2}.$$

The order in the numerator is: bottom times derivative of top, minus top times derivative of bottom. Reversing the two terms changes the sign of the answer, and it is a common error.

YOUR TURN 14.6 The Rules of Differentiation

27. Use the chain rule to differentiate each of the following.

(a) $y = (2x+1)^4$ [2] (b) $y = \sin(x^2)$ [2]

(c) $y = e^{3x}$ [2] (d) $y = \sqrt{3x+1}$ [3]

28. Use the product rule to differentiate each of the following.

(a) $y = x \cos x$ [2] (b) $y = x^2 e^x$ [2]
(c) $y = x^3 \ln x$ [3] (d) $y = x \sin x$ [2]

29. Use the quotient rule to differentiate each of the following.

(a) $y = \dfrac{x}{x+1}$ [2] (b) $y = \dfrac{x+1}{x-1}$ [3]
(c) $y = \dfrac{\sin x}{x}$ [3] (d) $y = \dfrac{e^x}{x}$ [3]

30. Explain, with a counterexample, why the derivative of a product is not the product of the derivatives. [3]

31. Find the gradient of $y = (x^2 - 1)^3$ at the point where $x = 2$. [3]

32. A student writes $\dfrac{d}{dx} \sin(3x) = \cos(3x)$. Identify the error and give the correct derivative. [2]

14.7 Higher Derivatives and Concavity

The derivative $f'(x)$ is itself a function, so it can be differentiated again. The result is the **second derivative** $f''(x)$, also written $\dfrac{d^2y}{dx^2}$.

If f' measures how fast f changes, then f'' measures how fast that *rate* changes. For motion this is easy to name: f is position, f' is velocity and f'' is acceleration.

You can continue. Differentiating n times gives the nth derivative, written $\dfrac{d^ny}{dx^n}$ or $f^{(n)}(x)$. You will need that notation for Maclaurin series. See Ch. 17.

Together, the signs of f' and f'' describe the shape of a graph completely. They answer two different questions, so keep them separate.

Sign	f': going up or down?	f'': bending which way?
positive	**increasing**	**concave up** (bends upward)
negative	**decreasing**	**concave down** (bends downward)
zero	**stationary point**: the curve is flat there	**point of inflexion**, *if* the concavity also changes

The two are independent. A curve can be increasing and concave down at the same time: it is rising, but by less and less. It can also be decreasing and concave up: it is falling, but by less and less.

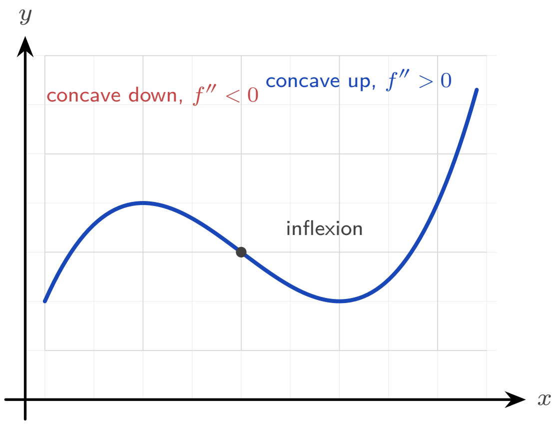

Figure 14.5 The second derivative decides which way the curve bends. Where the bending changes direction the curve has a point of inflexion.

EXAMPLE 14.13

For $f(x) = x^3 - 3x^2$, find $f'(x)$ and $f''(x)$, then state where f is increasing and where it is concave up.
Differentiate twice:

$$f'(x) = 3x^2 - 6x = 3x(x-2), \qquad f''(x) = 6x - 6 = 6(x-1).$$

f is increasing where $f'(x) > 0$, that is where $3x(x-2) > 0$, giving $x < 0$ or $x > 2$.
f is concave up where $f''(x) > 0$, that is $x > 1$.
At $x = 1$ the concavity changes from down to up: this is a **point of inflexion**, which you will study in Ch. 15.

Reading f, f' and f'' together

A common examination question shows one of these three graphs and asks about another. It helps to see all three at the same time.

Below are $f(x) = x^3 - 3x$ and its two derivatives, drawn one above the other on the same x-axis. The vertical scales are not the same. Only the zeros and the signs matter here.

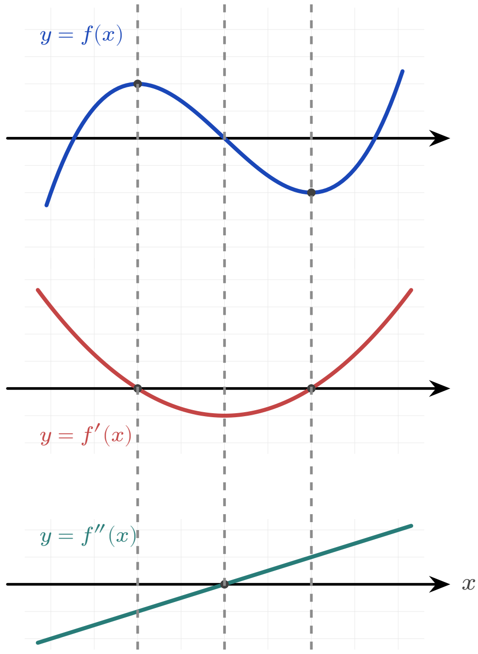

Figure 14.6 f, f' and f'' for $f(x) = x^3 - 3x$, stacked over a shared x-axis. Read down each dashed line: a turning point of f sits above a zero of f', and the inflexion of f sits above a turning point of f' and a zero of f''.

Read down each dashed line. At every dashed line, something happens in all three graphs at once, and the three results below are worth learning. Examinations test them in both directions: from f to f', and from f' back to f.

- A **turning point of** f sits above a **zero of** f'. At $x = -1$ the curve f has a maximum, and there f' crosses zero from positive to negative. At $x = 1$ the curve has a minimum, and there f' crosses zero from negative to positive.
- An **interval where** f **rises** sits above an **interval where** f' **is positive**, that is, where the graph of f' is above the axis.
- An **inflexion of** f sits above a **turning point of** f', and above a **zero of** f''. In this example all three occur at $x = 0$.

Use the table in either direction. Given any one of the three graphs, you can sketch the other two.

YOUR TURN 14.7 Higher Derivatives and Concavity

33. Find the second derivative of each of the following.

(a) $f(x) = x^3 - 6x^2$ [2]
(b) $y = x^4$ [2]
(c) $y = \sin x$ [2]
(d) $y = e^{2x}$ [2]

34. Find the interval or intervals on which each function is increasing.

(a) $f(x) = x^2 - 4x$ [2]
(b) $f(x) = x^4 - 2x^2$ [4]

35. Find the interval on which each curve is concave up.

(a) $y = x^3$ [2]
(b) $y = x^3 - 3x^2$ [3]

36. For $f(x) = x^3 - 3x^2$, find the coordinates of the point of inflexion and confirm that the concavity changes there. [4]

37. The graph of $y = f'(x)$ is a parabola with roots at $x = -2$ and $x = 3$, opening upwards. State the x-coordinates of the stationary points of f, and identify the nature of each. [4]

38. A particle moves so that its displacement after t seconds is $s = 2t^3 - 3t^2$ metres. Find expressions for the velocity and the acceleration, and find the time at which the acceleration is zero. [4]

14.8 L'Hôpital's Rule HL

This chapter began with the indeterminate form $\frac{0}{0}$, and solved it by factorising. Factorising fails as soon as the functions are trigonometric, exponential or logarithmic.

There is a second method for those cases. If a limit gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the derivative gives the answer. The rule is named after the French mathematician Guillaume de l'Hôpital, and the name is said *lo-pee-TAL*: the letter s is silent, and so is the h.

KEY **L'Hôpital's rule.** If $\lim_{x \to a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists.

Differentiate the top and the bottom *separately*. This is not the quotient rule. Then try the limit again, and if it is still indeterminate, apply the rule a second time.

Why the rule works HL

The reason is short, and it explains why the rule needs the $\frac{0}{0}$ form.

Suppose $f(a) = 0$ and $g(a) = 0$. Then near $x = a$ each function is close to its own tangent line at a, so

$$f(x) \approx f'(a)(x - a), \qquad g(x) \approx g'(a)(x - a).$$

Divide one by the other. The factor $(x - a)$ appears in both, so it cancels:

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x - a)}{g'(a)(x - a)} = \frac{f'(a)}{g'(a)}.$$

So the quotient of the functions approaches the quotient of their gradients. This is why both functions must vanish at a: only then does each one reduce to a multiple of $(x - a)$, and only then does the cancelling work.

EXAMPLE 14.14

Evaluate (a) $\lim_{x \to 0} \frac{\sin x}{x}$ and (b) $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$.

(a) Substituting gives $\frac{0}{0}$. Differentiate top and bottom:

$$\lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \frac{\cos x}{1} = \cos 0 = 1.$$

This confirms in one line the limit stated in §14.1.

(b) Substituting gives $\frac{0}{0}$. One application gives $\frac{\sin x}{2x}$, still $\frac{0}{0}$, so apply the rule again:

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \lim_{x \to 0} \frac{\sin x}{2x} = \lim_{x \to 0} \frac{\cos x}{2} = \frac{1}{2}.$$

CAUTION L'Hôpital's rule applies *only* to the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. Always check that the limit is indeterminate before differentiating. Applying it to a form like $\frac{2}{0}$ or $\frac{3}{5}$ gives a wrong answer.

EXAMPLE 14.15

Evaluate $\lim_{x \to \infty} \frac{\ln x}{x}$.
As $x \to \infty$ this is $\frac{\infty}{\infty}$. Differentiate top and bottom:

$$\lim_{x \to \infty} \frac{\ln x}{x} = \lim_{x \to \infty} \frac{1/x}{1} = \lim_{x \to \infty} \frac{1}{x} = 0.$$

The denominator increases without bound while the numerator increases very slowly, so the ratio approaches zero. L'Hôpital's rule makes that comparison exact.

YOUR TURN 14.8 L'Hôpital's Rule HL

39. Evaluate each limit, checking first that it is indeterminate.

(a) $\lim_{x \to 0} \frac{e^x - 1}{x}$ [2]

(b) $\lim_{x \to 0} \frac{\sin 2x}{x}$ [2]

(c) $\lim_{x \to 0} \frac{\tan x}{x}$ [2]

(d) $\lim_{x \to \infty} \frac{\ln x}{x^2}$ [2]

40. Each of these needs the rule applied twice. Evaluate them.

(a) $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ [3]

(b) $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$ [3]

41. Evaluate $\lim_{x \to \infty} \frac{x^2 + 1}{2x^2 - 3}$, first by L'Hôpital's rule and then by dividing every term by x^2. Confirm that the two methods agree. [4]

42. A student evaluates $\lim_{x \to 0} \frac{x + 2}{x + 1}$ by differentiating top and bottom, obtaining 1. Explain the error and give the correct value. [3]

43. State the two indeterminate forms to which L'Hôpital's rule applies, and explain why the rule must not be used on a form such as $\frac{2}{0}$. [3]

Chapter 14 · Limits & Derivatives

Reflections

TOK The early calculus was built on "infinitesimals," quantities treated as nonzero when dividing and as zero when convenient. The philosopher George Berkeley mocked them in 1734 as "ghosts of departed quantities," and he had a point: the logic was shaky. The method gave correct answers for 150 years before the limit gave it a rigorous foundation. Is a mathematical technique justified by its results, or only by its proof? Calculus worked long before anyone could say exactly why.

TOK A derivative gives a rate of change at a single instant, yet no instant can be measured: every measurement takes time. Is an instantaneous rate something the world contains, or a device invented to describe it? Does it matter, if the device predicts correctly every time?

TOK The expression $\frac{0}{0}$ has no value, and yet a limit of that form often has one exactly. Can a symbol that means nothing still carry information? Where does the meaning sit: in the symbols, or in the process that produced them?

IM Calculus was created twice, independently. Isaac Newton developed his "method of fluxions" in England in the 1660s; Gottfried Leibniz developed his version, with the $\frac{dy}{dx}$ notation we still use, in Germany in the 1670s. A bitter priority dispute split European mathematics for a century. Newton's notation was used in Britain and Leibniz's in the rest of Europe. British mathematics fell behind, partly because it kept the more awkward symbols. Notation is not neutral: the right symbol can speed up a whole field.

RW The derivative is the language of change, so it appears wherever anything varies. Velocity and acceleration in physics, marginal cost in economics, reaction rates in chemistry, growth rates in biology: each is a derivative. When a news report says inflation is "rising but more slowly," it is making a claim about a second derivative, that the rate of change of prices is itself decreasing.

EXT L'Hôpital's rule has a deeper cousin. Near a point, many functions are well approximated by a polynomial, the Maclaurin series: $\sin x \approx x - \frac{x^3}{6}$, $e^x \approx 1 + x + \frac{x^2}{2}$. Substituting these into an indeterminate limit resolves it without differentiating repeatedly. The same problem that L'Hôpital's rule solves leads on to representing functions as infinite polynomials, one of the great ideas of analysis.

End-of-Chapter Problems — Chapter 14: Limits & Derivatives

1. Differentiate $y = e^x \sin x$, and hence find the gradient of the curve at $x = 0$. [4]

2. Find the equation of the tangent to $y = x^2$ that is parallel to the line $y = 6x - 1$. [4]

3. Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{x}$. [4]

4. Differentiate $f(x) = x^3$ from first principles, showing every step of the limit. [4]

5. Using the quotient rule on $\tan x = \frac{\sin x}{\cos x}$, prove that $\frac{d}{dx} \tan x = \sec^2 x$. [4]

6. Consider $f(x) = x^3 - 3x$.

(a) Find $f'(x)$. [1]
(b) Find the coordinates of the points where the tangent is horizontal. [4]

7. The function $f(x) = \frac{e^x - 1}{x}$ is not defined at $x = 0$.

(a) Copy and complete the table of values of $f(x)$, giving 5 significant figures: $x = -0.1,\ -0.01,\ 0.01,\ 0.1$. [2]
(b) Suggest the value of $\lim_{x \to 0} f(x)$. [1]
(c) Verify your conjecture using l'Hôpital's rule. [2]

8. Differentiate $f(x) = 3x^2 - 2x$ from first principles. [5]

9. Consider $f(x) = |x - 2|$.

(a) Sketch the graph of f. [1]
(b) Explain briefly why f is continuous at $x = 2$. [2]
(c) Explain why f is not differentiable at $x = 2$. [2]

10. Evaluate $\lim_{x \to 0} \frac{1 - \cos 3x}{x^2}$. [5]

11. Find the coordinates of the points on the curve $y = x^3 - 3x$ where the tangent is parallel to the line $y = 9x + 1$. [5]

12. A curve has equation $y = \frac{e^{2x}}{1 + x^2}$.

(a) Find $\dfrac{dy}{dx}$. [4]
(b) Find the gradient of the curve at $x = 0$. [1]

13. Evaluate $\lim_{x \to 0} \dfrac{x - \sin x}{x^3}$, justifying each application of L'Hôpital's rule. [5]

14. Consider $f(x) = x^2 - 2x$.

(a) Find $f'(x)$ from first principles. [4]
(b) Hence find the gradient of the curve at $x = 3$. [2]

15. The curve $y = x^2 - 2x$ is considered at the point where $x = 3$.

(a) Find the coordinates of the point. [1]
(b) Find the equation of the tangent at this point. [3]
(c) Find the equation of the normal at this point. [2]

16. Evaluate the following limits.

(a) $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$ [3]
(b) $\lim_{x \to 0} \dfrac{e^x - 1 - x}{x^2}$ [3]

17. A curve has equation $y = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Find $\dfrac{dy}{dx}$. [3]
(b) Show that the curve has a horizontal tangent at $x = e$. [3]

18. Differentiate each of the following, simplifying your answers.

(a) $y = (2x^3 - 1)^4$ [2]
(b) $y = x^2 \ln x$ [2]
(c) $y = \dfrac{\sin x}{x}$ [2]

19. Consider the curve $y = e^x$ at the point where $x = 1$.

(a) Show that the tangent there passes through the origin. [4]
(b) Find the equation of the normal at the same point. [2]

20. Differentiate each of the following.

(a) $y = \arctan(3x)$ [2]
(b) $y = \sec(2x)$ [2]
(c) $y = \log_2 x$ [2]

21. Consider $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find the intervals on which f is increasing. [3]
(b) Find the interval on which f is concave up. [2]
(c) State the x-coordinate of the point of inflexion. [1]

22. Find the equations of *both* tangents to the curve $y = x^2$ that pass through the point $(0, -1)$. [6]

23. Differentiate each of the following, simplifying where possible.

(a) $y = (3x^2 + 1)^5$ [2]
(b) $y = x^2 e^x$ [2]
(c) $y = \dfrac{x}{x+1}$ [3]

24. Differentiate each of the following.

(a) $y = \sec x$ [1]
(b) $y = \ln(\cos x)$ [3]
(c) $y = x \arctan x$ [3]

25. The displacement of a particle is $s(t) = t^3 - 6t^2 + 9t$ metres at time t seconds, for $t \geq 0$.

(a) Find the velocity $v(t) = s'(t)$. [2]
(b) Find the times at which the particle is momentarily at rest. [3]
(c) Find the acceleration at $t = 3$. [2]

26. The derivative of a function f is $f'(x) = (x+2)(x-4)$.

(a) Find the intervals on which f is increasing. [2]
(b) Find the x-coordinates of the turning points of f and determine their nature. [3]
(c) Find the x-coordinate of the point of inflexion of f. [2]

27. Evaluate $\lim\limits_{x \to 0} \dfrac{\tan x - x}{x^3}$. You may use $\lim\limits_{x \to 0} \dfrac{\tan x}{x} = 1$ after establishing it. [7]

28. Consider $f(x) = x^3 - 3x^2 - 9x$.

(a) Find $f'(x)$ and the values of x where $f'(x) = 0$. [3]
(b) State the intervals on which f is increasing. [2]
(c) Find $f''(x)$ and the x-coordinate of the point of inflexion. [3]

29. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find $f'(x)$ and factorise it fully. [2]
(b) Find the coordinates of the two stationary points. [3]
(c) Find $f''(x)$, and use it to determine the nature of each stationary point. [3]
(d) Find the coordinates of the point of inflexion. [2]
(e) State the interval on which f is concave up. [2]

30. Let $g(x) = \dfrac{2x-1}{x+2}$, $x \neq -2$.

(a) Write down the equations of the vertical and horizontal asymptotes. [2]
(b) Use the quotient rule to show that $g'(x) = \dfrac{5}{(x+2)^2}$. [3]
(c) Hence explain why g is increasing on every interval of its domain. [2]
(d) Find the equation of the tangent to the curve at the point where $x = 0$. [2]
(e) Find the equation of the normal at the same point. [2]

31. Let $y = x^2e^{-x}$.

(a) Use the product rule to show that $\dfrac{dy}{dx} = x(2-x)e^{-x}$. [3]
(b) Find the coordinates of the two stationary points. [3]
(c) Show that $\dfrac{d^2y}{dx^2} = (x^2 - 4x + 2)e^{-x}$. [3]
(d) Use the second derivative to determine the nature of each stationary point. [2]
(e) Describe the behaviour of y as $x \to \infty$, and state the equation of the asymptote. [2]

32. GDC A closed cylindrical can has volume 500 cm^3. Its radius is r cm and its height is h cm.

(a) Show that $h = \dfrac{500}{\pi r^2}$. [2]
(b) Hence show that the total surface area is $S = 2\pi r^2 + \dfrac{1000}{r}$. [3]
(c) Find $\dfrac{dS}{dr}$. [2]
(d) Find the value of r that minimises S, giving your answer to three significant figures. [3]
(e) Verify, using the second derivative, that this value gives a minimum. [2]
(f) Show that at this radius the height is exactly twice the radius. [2]

Challenge Questions

33. The power rule, proved. Section 14.3 stated the power rule after two examples suggested it. Here you prove it for every positive integer n, using the binomial theorem.

(a) Write down the first three terms of $(x+h)^n$ in ascending powers of h. [3]

(b) Hence show that

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1},$$

and explain why every term after the first still contains h. [5]

(c) Deduce that $\frac{d}{dx}x^n = nx^{n-1}$ for every positive integer n. [3]

(d) Check your working by expanding in full for $n = 4$. [3]

(e) The rule holds for negative and fractional n as well, but this argument does not reach them. State which step fails when $n = -1$, then confirm that case by differentiating $\frac{1}{x}$ from first principles. [4]

34. Where the derivative of sine comes from. The standard derivatives in §14.5 rest on one limit. Here you establish that limit, then use it to prove the first of them.

(a) Evaluate $\frac{\sin x}{x}$ at $x = 0.1$, 0.01 and 0.001 radians, and at the matching negative values. Use the results as evidence that $\lim_{x\to 0}\frac{\sin x}{x} = 1$. [3]

(b) Show that $\frac{\cos x - 1}{x} = \frac{-\sin^2 x}{x(\cos x + 1)}$. Hence show that $\lim_{x\to 0}\frac{\cos x - 1}{x} = 0$. *(Hint: multiply above and below by $\cos x + 1$)* [5]

(c) The compound-angle identity gives $\sin(x+h) = \sin x\cos h + \cos x\sin h$. Use it to show that

$$\frac{\sin(x+h) - \sin x}{h} = \sin x\left(\frac{\cos h - 1}{h}\right) + \cos x\left(\frac{\sin h}{h}\right).$$

[4]

(d) Hence prove from first principles that $\frac{d}{dx}\sin x = \cos x$. [3]

(e) Explain why the proof fails if x is measured in degrees. [2]

35. Two routes to the same limit. An indeterminate limit can be settled by L'Hôpital's rule, or by replacing each function with its Maclaurin series. Here you use both on $\lim_{x\to 0}\frac{x - \sin x}{x^3}$ and compare them.

(a) Evaluate the limit by applying L'Hôpital's rule repeatedly, checking at each stage that the form is still indeterminate. [5]

(b) The Maclaurin series for $\sin x$ begins $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$. Substitute it into the fraction and simplify, and the limit follows in one step. [4]

(c) Use the series method on $\lim_{x\to 0}\frac{e^x - 1 - x}{x^2}$, given that $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$. [3]

(d) Give one advantage of each method, and one limit for which the series method would be harder to use. [4]